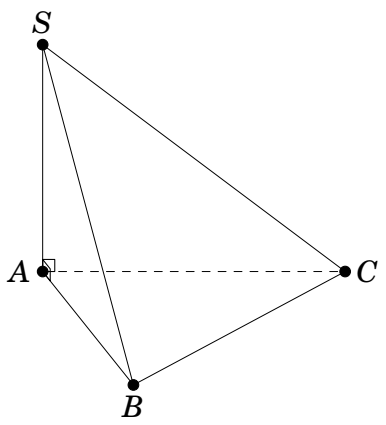
BỘ KHUNG MẪU HÌNH VẼ TIKZ CHÍNH THỨC (BASE TOPOLOGY)

Phục vụ render động trên Streamlit Web App

A. NHÓM CHÓP TAM GIÁC SERIES

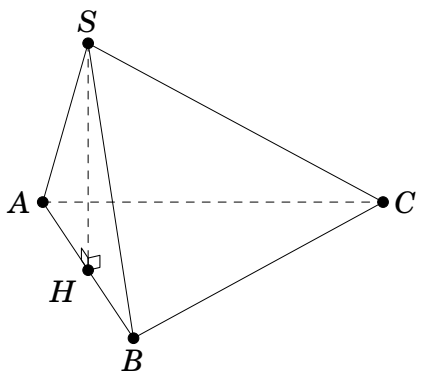
A.1. Chóp tam giác - Cạnh bên vuông góc đáy (SA ⊥ ABC)



Code TikZ Base

```
\begin{tikzpicture}[scale=1,>=stealth,
    font=\footnotesize, line join=round, line cap=round]
  \tkzDefPoints{0/0/A,1.2/-1.5/B,4/0/C}
  \coordinate (S) at ($(A)+(0,3)$);
  \tkzDrawPolygon(S,A,B,C)
  \tkzDrawSegments(S,B)
  \tkzDrawSegments[dashed](A,C)
  \tkzDrawPoints[fill=black,size=4](A,B,C,S)
  \tkzMarkRightAngles[size=0.16](S,A,B S,A,C)
  \tkzLabelPoints[above](S)
  \tkzLabelPoints[below](B)
  \tkzLabelPoints[left](A)
  \tkzLabelPoints[right](C)
\end{tikzpicture}
```

A.2. Chóp tam giác - Mặt bên vuông góc đáy (SAB ⊥ ABC)

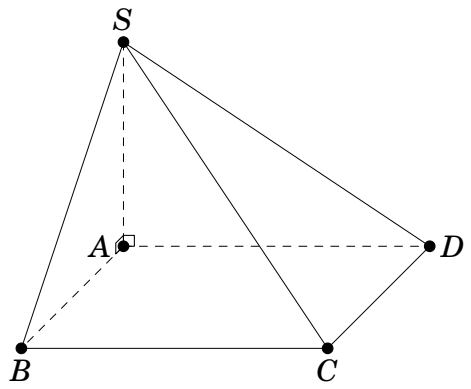


Code TikZ Base

```
\begin{tikzpicture}[scale=1,>=stealth,...]
  \tkzDefPoints{0/0/A,1.2/-1.8/B,4.5/0/C}
  \coordinate (H) at ($(A)!0.5!(B)$);
  \coordinate (S) at ($(H)+(0,3)$);
  \tkzDrawPolygon(S,A,B,C)
  \tkzDrawSegments(S,B)
  \tkzDrawSegments[dashed](A,C S,H)
  \tkzDrawPoints[fill=black,size=4](A,B,C,S,H)
  \tkzMarkRightAngles[size=0.16](S,H,A S,H,C)
  \tkzLabelPoints[above](S)
  \tkzLabelPoints[left](A) ...
\end{tikzpicture}
```

B. NHÓM CHÓP TỨ GIÁC SERIES

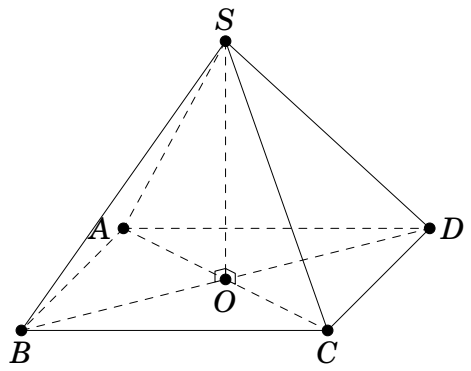
B.1. Chóp tứ giác đáy HBH - Cạnh bên vuông góc đáy (SA ⊥ ABCD)



Code TikZ Base

```
\begin{tikzpicture}[scale=0.9,>=stealth,...]
  \tkzDefPoints{0/0/A, -1.5/-1.5/B, 3/-1.5/C}
  \coordinate (D) at ($(A)+(C)-(B)$);
  \coordinate (S) at ($(A)+(0,3)$);
  \tkzDrawPolygon(S,B,C,D)
  \tkzDrawSegments(S,C)
  \tkzDrawSegments[dashed](S,A A,B A,D)
  \tkzDrawPoints[fill=black,size=4](A,B,C,D,S)
  \tkzMarkRightAngles[size=0.16](S,A,B S,A,D)
  \tkzLabelPoints[above](S) ...
\end{tikzpicture}
```

B.2. Chóp tứ giác đều (SO ⊥ ABCD)

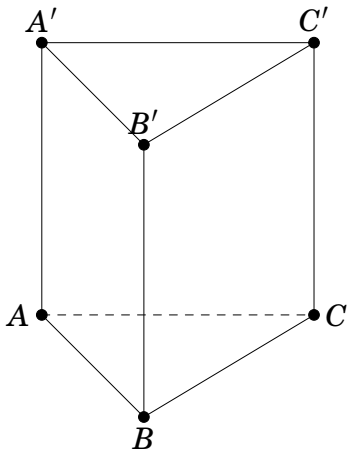


Code TikZ Base

```
\begin{tikzpicture}[scale=0.9,>=stealth,...]
  \tkzDefPoints{0/0/A, -1.5/-1.5/B, 3/-1.5/C}
  \coordinate (D) at ($(A)+(C)-(B)$);
  \coordinate (O) at ($(A)!0.5!(C)$);
  \coordinate (S) at ($(O)+(0,3.5)$);
  \tkzDrawPolygon(S,B,C,D)
  \tkzDrawSegments(S,C)
  \tkzDrawSegments[dashed](S,A A,B A,D
                           A,C B,D S,O)
  \tkzDrawPoints[fill=black,size=4](A,B,C,D,S,O)
  \tkzMarkRightAngles[size=0.16](S,O,B S,O,C)
\end{tikzpicture}
```

C. NHÓM LĂNG TRỤ & HÌNH HỘP SERIES

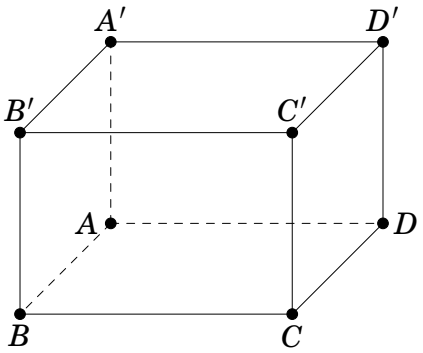
C.1. Lăng trụ đứng tam giác



Code TikZ Base

```
\begin{tikzpicture}[scale=0.9,>=stealth,...]
  \tkzDefPoints{0/0/A, 1.5/-1.5/B, 4/0/C}
  \coordinate (A') at ($(A)+(0,4)$);
  \coordinate (B') at ($(B)+(0,4)$);
  \coordinate (C') at ($(C)+(0,4)$);
  \tkzDrawPolygon(A',B',C')
  \tkzDrawSegments(A,A' B,B' C,C' A,B B,C)
  \tkzDrawSegments[dashed](A,C)
  \tkzDrawPoints[fill=black,size=4](A,B,C,A',B',C')
  \tkzLabelPoints[above](A',B',C')
\end{tikzpicture}
```

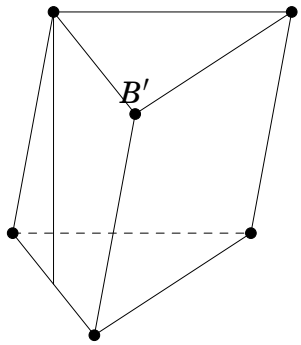
C.2. Hình hộp chữ nhật



Code TikZ Base

```
\begin{tikzpicture}[scale=0.8,>=stealth,...]
  \tkzDefPoints{0/0/A, -1.5/-1.5/B, 3/-1.5/C}
  \coordinate (D) at ($(A)+(C)-(B)$);
  \coordinate (A') at ($(A)+(0,3)$);
  ... Tương tự tịnh tiến B', C', D'
  \tkzDrawPolygon(A',B',C',D')
  \tkzDrawSegments(B,C C,D B,B' C,C' D,D')
  \tkzDrawSegments[dashed](A,B A,D A,A')
  \tkzDrawPoints[fill=black,size=4](A,B,C...
\end{tikzpicture}
```

C.3. Lăng trụ xiên tam giác

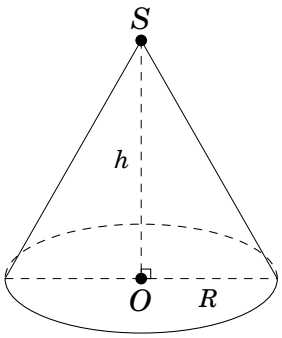


Code TikZ Base

```
\begin{tikzpicture}[scale=0.9,>=stealth,...]
  \tkzDefPoints{0/0/A,1.2/-1.5/B,3.5/0/C}
  \coordinate (H) at ($(A)!1/2!(B)$);
  \coordinate (A') at ($(H)+(0,4)$);
  \tkzDefPointsBy[translation = from A to A'](B,C){B'}{C'}
  \tkzDrawPolygon(A,B,C,C',B',A')
  \tkzDrawSegments(A',C' B',B A',H)
  \tkzDrawSegments[dashed](A,C)
\end{tikzpicture}
```

D. NHÓM KHỐI TRÒN XOAY SERIES

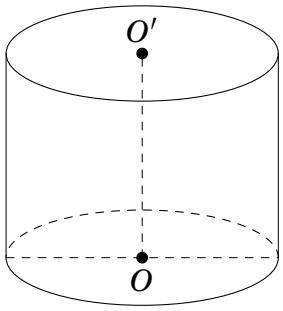
D.1. Hình Nón



Code TikZ Base

```
\begin{tikzpicture}[scale=0.9,>=stealth,...]
  \def \x{2} \def \y{0.8} \def \h{3.5}
  \coordinate (O) at (0,0);
  \coordinate (S) at ($(O)+(0,\h)$);
  \coordinate (l) at (180:\x cm and \y cm);
  \coordinate (r) at (0:\x cm and \y cm);
  \draw[dashed] (r) arc (0:180:\x cm and \y cm);
  \draw (r) arc (0:-180:\x cm and \y cm);
  \tkzDrawSegments(S,l S,r)
  \tkzDrawSegments[dashed](S,O l,r)
\end{tikzpicture}
```

D.2. Hình Trụ



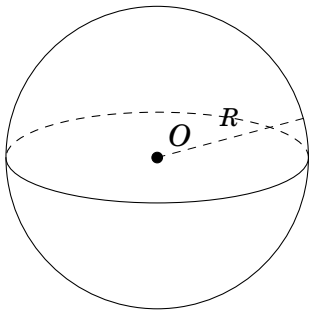
Code TikZ Base

```
\begin{tikzpicture}[scale=0.9,>=stealth,...]
  \def \x{2} \def \y{0.7} \def \h{3}
  \coordinate (O) at (0,0);
  \coordinate (O') at ($(O)+(0,\h)$);
  \coordinate (Al) at (180:\x cm and \y cm); ...

  \draw[dashed] (Ar) arc (0:180:\x cm and \y cm);
  \draw (Ar) arc (0:-180:\x cm and \y cm);
  \draw (A'r) arc (0:360:\x cm and \y cm);

  \tkzDrawSegments(Al,A'l Ar,A'r)
  \tkzDrawSegments[dashed](O,O' Al,Ar)
\end{tikzpicture}
```

D.3. Hình Cầu

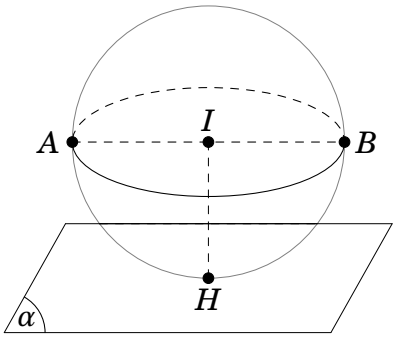


Code TikZ Base

```
\begin{tikzpicture}[scale=1,>=stealth,...]
  \coordinate (O) at (0,0);
  \def\R{2}
  \draw (O) circle (\R);
  \draw[dashed] (\R,0) arc
         (0:180:\R cm and 0.3*\R cm);
  \draw (\R,0) arc
         (0:-180:\R cm and 0.3*\R cm);
  \tkzDrawPoints[fill=black,size=4](O)
  \tkzLabelPoints[above right](O)
  \coordinate (A) at (15:\R);
  \tkzDrawSegments[dashed](O,A)
\end{tikzpicture}
```

E. NHÓM VỊ TRÍ TƯƠNG ĐỐI (MẶT CẦU & MẶT PHẲNG)

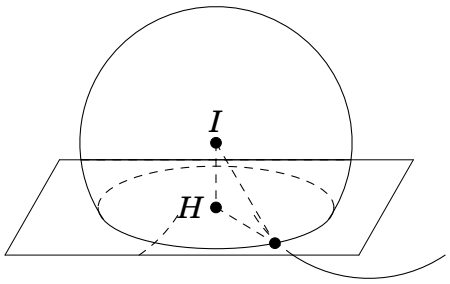
E.1. Mặt cầu tiếp xúc Mặt phẳng



Code TikZ Base

```
\begin{tikzpicture}[scale=0.9,>=stealth,...]
  \def \x{2} \def \y{0.8}
  \coordinate (I) at (0,0);
  \coordinate (B) at (\x,0);
  \coordinate (A) at (-\x,0);
  \coordinate (H) at (-90:\x);
  \coordinate (m) at ($(H)+(-\x-1,-\y)$); ...
  \tkzDrawCircle[radius](I,A)
  \draw[dashed] (A)--(B) arc (0:180:\x cm and \y cm);
  \draw (B) arc (0:-180:\x cm and \y cm);
  \tkzInterLC(p,q)(I,A)\tkzGetPoints{p'}{q'}
  \tkzDrawSegments[dashed](p',q' I,H)
  \tkzDrawSegments(q',q q,m m,n n,p p,p')
  \tkzMarkAngles[size=0.6cm,arc=l](n,m,q)
\end{tikzpicture}
```

E.2. Mặt cầu cắt Mặt phẳng

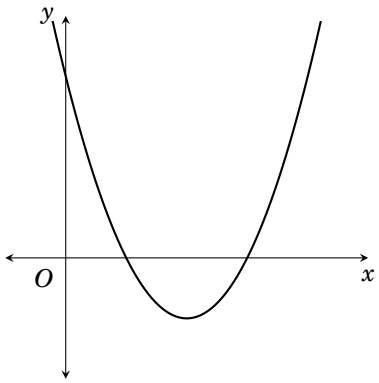


Code TikZ Base

```
\begin{tikzpicture}[scale=0.9,>=stealth,...]
  \def \x{2} \def \y{0.7}
  \coordinate (I) at (0,0);
  \coordinate (B) at (\x,0); ...
  \tkzInterLC(p,q)(I,A)\tkzGetPoints{p'}{q'}
  \tkzInterLC(m,n)(I,A)\tkzGetPoints{x}{y}
  \draw (-34:\x) arc (-34:214:\x);
  \draw (y) arc (-124:-56:\x);
  \draw (x)[dashed] arc (-56:-30:\x);
  \draw (y)[dashed] arc (-124:-150:\x);
  \draw[dashed] ($(-30:\x)+(0,0.05)$) arc ...
  \draw ($(-30:\x)+(0,0.05)$) arc ...
  \draw[dashed] (p')--(q') (I)--(H)...
\end{tikzpicture}
```

F. NHÓM ĐỒ THỊ HÀM SỐ & HỆ TRỤC OXY

F.1. Đồ thị hàm Parabol (Bậc 2)

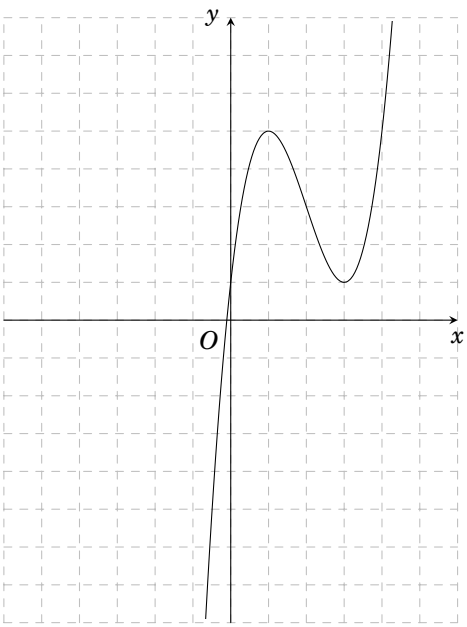


Code TikZ Base

```
\begin{tikzpicture}[scale=0.8,>=stealth,...]
  \def\a{1} \def\b{-4} \def\c{3}
  \def\xmin{-1} \def\xmax{5}
  \def\ymin{-2} \def\ymax{4}

  \draw[<->] (\xmin,0)--(\xmax,0) node [below]{$x$};
  \draw[<->] (0,\ymin)--(0,\ymax) node [left]{$y$};
  \node at (0,0) [below left]{$O$};
  \clip (\xmin+0.1,\ymin+0.1) rectangle ...
  \draw[smooth,samples=100, thick]
    plot(\x,{\a*(\x)^2+\b*(\x)+\c});
\end{tikzpicture}
```

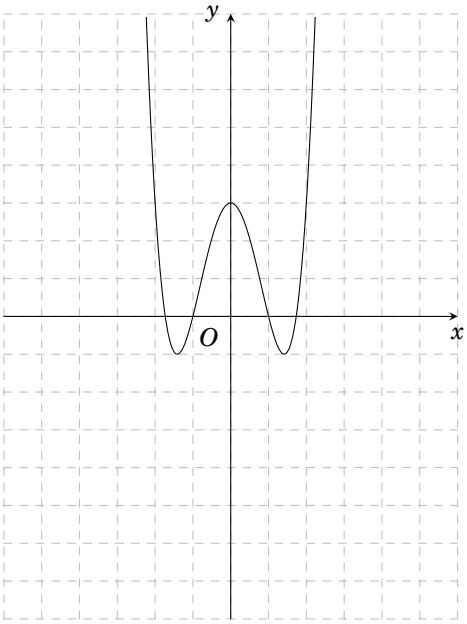
F.2. Đồ thị hàm số bậc 3



Code TikZ Base

```
\begin{tikzpicture}[scale=0.5,>=stealth,...]
  \def\a{1} \def\b{-6} \def\c{9} \def\d{1}
  \def\xmin{-6} \def\xmax{6}
  \def\ymin{-8} \def\ymax{8}
  \draw[color=gray!50,dashed] (\xmin,\ymin) grid (\xmax,\ymax);
  \draw[->] (\xmin,0)--(\xmax,0) node [below]{$x$};
  \draw[->] (0,\ymin)--(0,\ymax) node [left]{$y$};
  \node at (0,0) [below left]{$O$};
  \clip (\xmin+0.1,...);
  \draw[smooth,samples=100] plot(\x,{\a*(\x)^3+\b*(\x)^2+\c*(\x)+\d});
\end{tikzpicture}
```

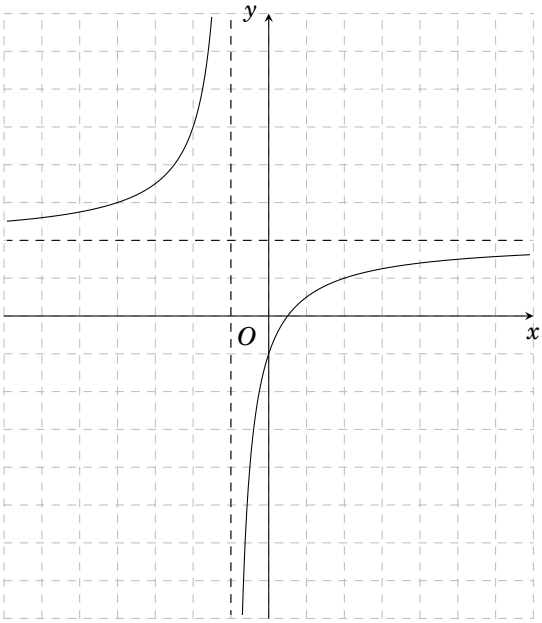
F.3. Đồ thị hàm số trùng phương



Code TikZ Base

```
\begin{tikzpicture}[scale=0.5,>=stealth,...]
  \def\a{1} \def\b{-4} \def\c{3}
  \def\xmin{-6} \def\xmax{6}
  \def\ymin{-8} \def\ymax{8}
  \draw[color=gray!50,dashed] (\xmin,\ymin) grid (\xmax,\ymax);
  \draw[->] (\xmin,0)--(\xmax,0) node [below]{$x$};
  \draw[->] (0,\ymin)--(0,\ymax) node [left]{$y$};
  \node at (0,0) [below left]{$O$};
  \clip (\xmin+0.1,...);
  \draw[smooth,samples=100] plot(\x,{\a*(\x)^4+\b*(\x)^2+\c});
\end{tikzpicture}
```

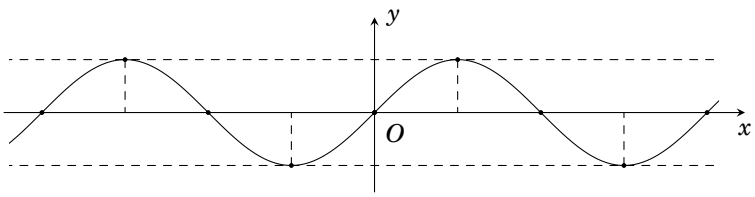
F.4. Đồ thị hàm số Nhất biến (Phân thức $\frac{ax+b}{cx+d}$)



Code TikZ Base

```
\begin{tikzpicture}[scale=0.5,>=stealth,...]
  \def\a{2} \def\b{-1} \def\c{1} \def\d{1}
  \def\xmin{-7} \def\xmax{7}
  \def\ymin{-8} \def\ymax{8}
  % Vẽ lưới và trục
  \clip (\xmin+0.1,\ymin+0.1) rectangle (\xmax-0.1,\ymax-0.1);
  % Vẽ miền x < -d/c
  \draw[smooth,domain=\xmin:(-\d/\c-0.1)] plot(\x,{(\a*(\x)+\b)/(\c*(\x
  % Vẽ miền x > -d/c
  \draw[smooth,domain=(-\d/\c+0.1:\xmax)] plot(\x,{(\a*(\x)+\b)/(\c*(\x
  % Tiệm cận đứng và ngang
  \draw[dashed] (-\d/\c,\ymin)--(-\d/\c,\ymax);
  \draw[dashed] (\xmin,\a/\c)--(\xmax,\a/\c);
\end{tikzpicture}
```

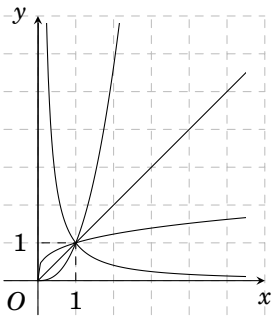
F.5. Đồ thị hàm lượng giác (Sin)



Code TikZ Base

```
\begin{tikzpicture}[scale=0.7,>=stealth,...]
  \def\xmin{-7} \def\xmax{7} \def\ymin{-1.5} \def\ymax{1.8}
  \draw[->] (\xmin,0)--(\xmax,0) node [below]{$x$};
  \draw[->] (0,\ymin)--(0,\ymax) node [right]{$y$};
  \node at (0,0) [below right]{$O$};
  \clip (\xmin+0.1,\ymin+0.1) rectangle (\xmax-0.5,\ymax-0.1);
  \draw[smooth,samples=200,domain=\xmin:\xmax] plot(\x,{sin(\x r)});
  \draw[dashed] (\xmin,1)--(\xmax,1) (\xmin,-1)--(\xmax,-1);
  % Vẽ các điểm đặc biệt
  \foreach \x in {-2*pi,-1.5*pi,-pi,-0.5*pi,0}
  {\draw[fill=black] (\x,sin \x*180/pi) circle (1pt);
   \draw[dashed] (\x,sin \x*180/pi)--(\x,0);
   \draw[fill=black] (-\x,sin -\x*180/pi) circle (1pt);
   \draw[dashed] (-\x,sin -\x*180/pi)--(-\x,0);}
\end{tikzpicture}
```

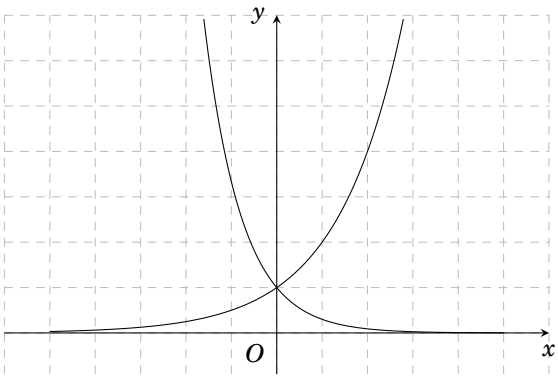
F.6. Đồ thị hàm số Lũy thừa & Logarit



Code TikZ Base

```
\begin{tikzpicture}[scale=0.5,>=stealth,...]
  \def\xmax{6} \def\ymax{7}
  \draw[color=gray!50,dashed] (-0.9,-0.9) grid (\xmax,\ymax);
  \draw[->] (-0.9,0)--(\xmax,0) node [below]{$x$};
  \draw[->] (0,-0.9)--(0,\ymax) node [left]{$y$};
  \clip (-0.9,-0.9) rectangle (\xmax-0.5,\ymax-0.2);
  % Vẽ đường y = x^a theo các trường hợp a
  \draw[smooth,domain=0:\xmax] plot(\x,{(\x)^2.5});
  \draw[smooth,domain=0:\xmax] plot(\x,{\x});
  \draw[smooth,domain=0:\xmax] plot(\x,{(\x)^0.3});
  \draw[smooth,domain=0:\xmax] plot(\x,{(\x)^(-1.3)});
  \draw[dashed] (1,0) node[below]{$1$}--(1,1)--(0,1)node[left]{$1$};
\end{tikzpicture}
```

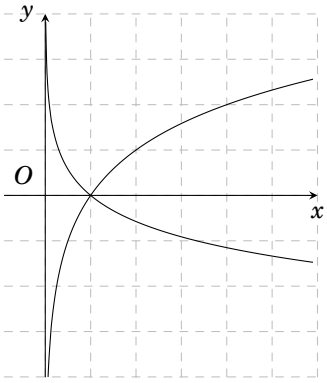
F.7. Đồ thị hàm số Mũ (a^x)



Code TikZ Base

```
\begin{tikzpicture}[scale=0.6,>=stealth,...]
  \def\xmin{-6} \def\xmax{6} \def\ymax{7}
  \draw[color=gray!50,dashed] (\xmin,-0.9) grid (\xmax,\ymax);
  \draw[->] (\xmin,0)--(\xmax,0) node [below]{$x$};
  \draw[->] (0,-0.9)--(0,\ymax) node [left]{$y$};
  \clip (\xmin,-0.9) rectangle (\xmax-0.1,\ymax-0.1);
  \draw[smooth,samples=100] plot(\x,{2^(\x)});
  \draw[smooth,samples=100] plot(\x,{0.3^(\x)});
\end{tikzpicture}
```

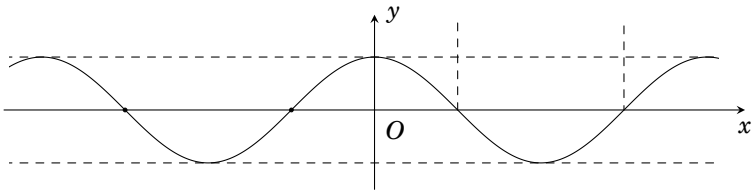
F.8. Đồ thị hàm số Logarit ($\log_a x$)



Code TikZ Base

```
\begin{tikzpicture}[scale=0.6,>=stealth,...]
  \def\xmax{6} \def\ymin{-4} \def\ymax{4}
  \draw[color=gray!50,dashed] (-0.9,\ymin) grid (\xmax,\ymax);
  \draw[->] (-0.9,0)--(\xmax,0) node [below]{$x$};
  \draw[->] (0,\ymin)--(0,\ymax) node [left]{$y$};
  \clip (-0.9,\ymin) rectangle (\xmax-0.1,\ymax-0.1);
  \draw[smooth,samples=300,domain=0.01:\xmax] plot(\x,{ln(\x)/ln(2)});
\end{tikzpicture}
```

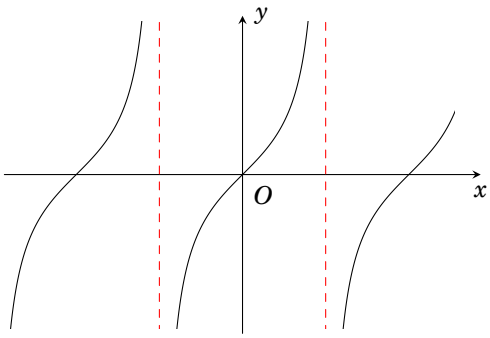
F.9. Đồ thị hàm lượng giác (Cosin)



Code TikZ Base

```
\begin{tikzpicture}[scale=0.7,>=stealth,...]
  \def\xmin{-7} \def\xmax{7} \def\ymin{-1.5} \def\ymax{1.8}
  \draw[->] (\xmin,0)--(\xmax,0) node [below]{$x$};
  \draw[->] (0,\ymin)--(0,\ymax) node [right]{$y$};
  \node at (0,0) [below right]{$O$};
  \clip (\xmin+0.1,\ymin+0.1) rectangle (\xmax-0.5,\ymax-0.1);
  \draw[smooth,samples=200,domain=\xmin:\xmax] plot(\x,{cos(\x r)});
  \draw[dashed] (\xmin,1)--(\xmax,1) (\xmin,-1)--(\xmax,-1);
  \foreach \x in {-1.5*pi,-0.5*pi,0.5*pi,1.5*pi}
  {\draw[fill=black] (\x,cos \x*180/pi) circle (1pt);
   \draw[dashed] (\x,cos \x*180/pi)--(\x,0);}
\end{tikzpicture}
```

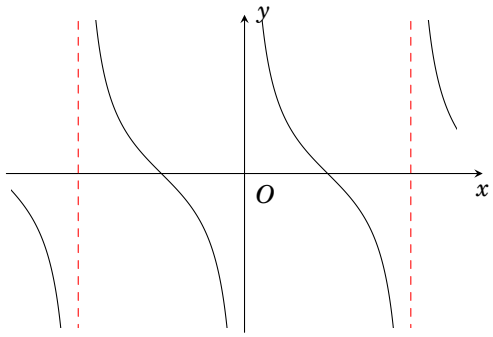
F.10. Đồ thị hàm lượng giác (Tan)



Code TikZ Base

```
\begin{tikzpicture}[scale=0.7,>=stealth,...]
  \def\xmin{-4.5} \def\xmax{4.5} \def\ymin{-3} \def\ymax{3}
  \draw[->] (\xmin,0)--(\xmax,0) node [below]{$x$};
  \draw[->] (0,\ymin)--(0,\ymax) node [right]{$y$};
  \node at (0,0) [below right]{$O$};
  \clip (\xmin+0.1,\ymin+0.1) rectangle (\xmax-0.5,\ymax-0.1);
  \draw[smooth,samples=200,domain=-1.4:1.4] plot(\x,{tan(\x r)});
  \draw[smooth,samples=200,domain=1.7:4.5] plot(\x,{tan(\x r)});
  \draw[smooth,samples=200,domain=-4.5:-1.7] plot(\x,{tan(\x r)});
  \foreach \x in {-1.5*pi,-0.5*pi,0.5*pi,1.5*pi}
  {\draw[dashed, color=red] (\x,\ymin)--(\x,\ymax);}
\end{tikzpicture}
```

F.11. Đồ thị hàm lượng giác (Cotan)



Code TikZ Base

```
\begin{tikzpicture}[scale=0.7,>=stealth,...]
  \def\xmin{-4.5} \def\xmax{4.5} \def\ymin{-3} \def\ymax{3}
  \draw[->] (\xmin,0)--(\xmax,0) node [below]{$x$};
  \draw[->] (0,\ymin)--(0,\ymax) node [right]{$y$};
  \node at (0,0) [below right]{$O$};
  \clip (\xmin+0.1,\ymin+0.1) rectangle (\xmax-0.5,\ymax-0.1);
  \draw[smooth,samples=200,domain=-3:-0.14] plot(\x,{cot(\x r)});
  \draw[smooth,samples=200,domain=0.14:3] plot(\x,{cot(\x r)});
  \draw[smooth,samples=200,domain=3.28:4.5] plot(\x,{cot(\x r)});
  \draw[smooth,samples=200,domain=-4.5:-3.28] plot(\x,{cot(\x r)});
  \foreach \x in {-1*pi,pi}
  {\draw[dashed, color=red] (\x,\ymin)--(\x,\ymax);}
\end{tikzpicture}
```

G. NHÓM BẢNG BIẾN THIÊN & BẢNG XÉT DẤU (TKZ-TAB)

G.1. Bảng biến thiên (Bậc 3 - 2 cực trị)

x	$-\infty$		x_1		x_2		$+\infty$
y'		$+$	0	$-$	0	$+$	
y	$-\infty$	$\nearrow$	y_1	$\searrow$	y_2	$\nearrow$	$+\infty$

Code TikZ Base

```
\begin{tikzpicture}[scale=1]
\tkzTabInit[nocadre=false,...] % Setup khung
  {$x$ /0.8, $y'$ /0.8, $y$ /2}
  {$-\infty$,$x_1$,$x_2$,$+\infty$}
  \tkzTabLine{,+,$0$,-,$0$,+,}
  \tkzTabVar{-/$-\infty$,+/$y_1$,-/$y_2$,+/$+\infty$}
\end{tikzpicture}
```

G.2. Bảng xét dấu 1 dòng

x	$-\infty$		-2		1		$+\infty$
$f(x)$		$+$	0	$-$	0	$+$	

Code TikZ Base

```
\begin{tikzpicture}[scale=1]
\tkzTabInit[nocadre=false,lgt=1.5,espcl=1.5,...]
  {$x$ /0.8,$f(x)$ /0.8}
  {$-\infty$,$-2$,$1$,$+\infty$}
  \tkzTabLine{,+,$0$,-,$0$,+,}
\end{tikzpicture}
```

G.3. Bảng biến thiên (Bậc 3 - Vô nghiệm)

x	$-\infty$		$+\infty$
y'		$-$	
y	$+\infty$	$\searrow$	$-\infty$

Code TikZ Base

```
\begin{tikzpicture}[scale=1]
\tkzTabInit[nocadre=false,...]
  {$x$/0.6,$y'$/0.6,$y$/2}
  {$-\infty$,$+\infty$}
  \tkzTabLine{,-,}
  \tkzTabVar{+/$+\infty$,-/$-\infty$}
\end{tikzpicture}
```

G.4. Bảng biến thiên (Hàm Nhất biến - Phân thức)

x	$-\infty$		x_0		$+\infty$
y'		$+$	‖	$+$	
y	y_0	$\nearrow$	$+\infty$ ‖ $-\infty$	$\nearrow$	y_0

Code TikZ Base

```
\begin{tikzpicture}[scale=1]
\tikzset{double style/.append style={
   double distance=1.5pt}}
\tkzTabInit[nocadre=false,...]
  {$x$ /0.8,$y'$ /0.8,$y$ /2}
  {$-\infty$,$x_0$,$+\infty$}
  % Dùng ký hiệu 'd' (double line)
  \tkzTabLine{,+,d,+,}
  \tkzTabVar{-/$y_0$,+D-/$+\infty$/$-\infty$,+/$y_0$}
\end{tikzpicture}
```

G.5. Bảng xét dấu 3 dòng

x	$-\infty$		-2		1		3		$+\infty$
x^2-x+6		$+$	0	$-$	|	$-$	0	$+$	
$x-1$		$-$	|	$-$	0	$+$	|	$+$	
$\frac{x^2-x-6}{x-1}$		$-$	0	$+$	||	$-$	0	$+$	

Code TikZ Base

```
\begin{tikzpicture}[scale=0.9]
\tkzTabInit[nocadre=false,lgt=2.5,...]
  {$x$, $x^2-x+6$, $x-1$, $\frac{...}{...}$}
  {$-\infty$,$-2$,$1$,$3$,$+\infty$}
  \tkzTabLine{,+,$0$,-,|,-,$0$,+,}
  \tkzTabLine{,-,|,-,$0$,+,|,+,}
  % Dùng ký tự || cho không xác định
  \tkzTabLine{,-,$0$,+,||,-,$0$,+,}
\end{tikzpicture}
```